AI Report

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Simulasi Monte Carlo.docx - 10/22/2026

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Matkul : Simulasi dan Pemodelan (Simulasi Monte Carlo)

SIMULASI MONTE CARLO UNTUK MEMPERKIRAKAN JUMLAH ITEM YANG TERJUAL PADA SEBUAH TOKO ROTI MENGGUNAKAN BAHASA R

Deskripsi Kasus:

Jumlah roti yang terjual per hari pada sebuah toko roti (French Bakery) tidak selalu tetap karena jumlah pembelian bisa berubah-ubah.
Untuk memperkirakan penjualan tersebut, digunakan metode Simulasi Monte Carlo dengan memanfaatkan data penjualan sebelumnya.

Simulasi ini bertujuan untuk mengetahui rata-rata penjualan harian dan melihat kemungkinan total penjualan selama 30 hari.

Sumber Data dan Bahasa Pemrograman

Sumber: https://www.kaggle.com/datasets/matthieugimbert/french-bakery-daily-sales

Dataset awal dianalisis menggunakan Python pada Google Colab untuk memperoleh total penjualan harian sebelum disimulasikan menggunakan bahasa R.

Dari total 600 catatan pengamatan penjualan harian, tercatat nilai kuartil pertama (Q1) sebesar 366 item serta kuartil ketiga (Q3) di angka 780,75 item.
Selanjutnya, data dikelompokkan menjadi tiga kondisi penjualan:

Kondisi

Frekuensi

Rata-rata Item Terjual

Sepi

149 Hari

306,10

Sedang

301 Hari

511,98

Ramai

150 Hari

1.068,49

Total

600 Hari

Metode Simulasi Monte Carlo

Langkah-langkah yang dilakukan dalam simulasi sebagai berikut:

Menentukan distribusi probabilitas

Menghitung probabilitas kumulatif

Menentukan interval bilangan acak 1-100

Melakukan 1.000 percobaan Monte Carlo

Melakukan 1.000 skenario penjualan selama 30 hari.

Berikut adalah tabel probabilitas, probabilitas kumulatif, dan interval bilangan acak yang telah dibulatkan:

Kondisi

Probabilitas

Probabilitas Kumulatif

Interval Acak

Sepi

0,25

0,25

1 - 25

Sedang

0,50

0,75

26 - 75

Ramai

0,25

1,00

76 - 100

Hasil Simulasi

Setelah dilakukan 1.000 percobaan simulasi, hasil yang diperoleh adalah:

Kondisi

Hasil

Rata-rata simulasi

602,69 item/hari

Rata-rata teoritis

599,64 item/hari

Selisih

0,51%

Hasil simulasi Monte Carlo menunjukkan rata-rata penjualan harian sebesar 602,69 item, sedangkan hasil perhitungan teoritis sebesar 599,64 item.
Selisih keduanya sekitar 0,51%, sehingga hasil simulasi cukup mendekati nilai teoritis.

Grafik Frekuensi Kondisi Penjualan & Total Penjualan Selama 30 Hari

Kesimpulan

Simulasi Monte Carlo menghasilkan rata-rata penjualan harian sebesar 602,69 item.
Angka ini sangat dekat dengan rata-rata teoritis sebesar 599,64 item, dengan selisih hanya 0,51%.
Untuk periode 30 hari, jumlah penjualan yang dihasilkan simulasi berbeda-beda pada setiap percobaan, tetapi umumnya berada di sekitar 18.000 item.
Hasil tersebut mendekati perhitungan teoritis sebesar 17.989,11 item.
Berdasarkan hasil tersebut, metode Monte Carlo dapat digunakan untuk memperkirakan jumlah penjualan roti dengan mempertimbangkan kemungkinan perubahan jumlah penjualan setiap harinya.

High Human Impact High AI Impact

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 20 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.